Statistics

Cse 33

Name: _____ Class Roll: _____ CT #01

দেখি সব পারি কি না

Q1. A food-delivery company analyzes the delivery time of 15 orders: [10 points]

18, 19, 20, 20, 21, 21, 21, 22, 23, 24, 25, 26, 28, 35, 65

Perform an exploratory analysis by:

a) Calculating mean, median and mode.

b) Identifying which statistic best represents a typical delivery.

c) Identifying any potential outlier.

d) Explaining how the outlier might affect a predictive model.

e) Draw box plot

Q2. A data scientist compares the box plots of execution times (in milliseconds) of four search algorithms: **Linear Search (LS), Binary Search (BS), Jump Search (JS), and Exponential Search (ES). [10 points]**

Metric	LS	BS	JS	ES
Q1	40	10	18	8
Median	55	14	24	10
Q3	70	18	30	12
Max	200	22	36	14
Min	30	8	15	7

a) Which algorithm has the most consistent (stable) performance? Explain.

b) Which algorithm is likely best suited for real-time applications requiring low-latency (time required to produce the output) and stable performance? Explain.

c) Find if there is any outlier in LS? If yes, what could be the possible interpretation of the performance of LS algorithm?

Q3. Derive batch gradient descent algorithm to solve linear regression problem. Write batch gradient descent algorithm. Mention the differences between batch, mini-batch and stochastic gradient descent algorithm. [10 points]

Name:_____ Class Roll:_____ CT#02

দেখি সব পারি কি না

Q1. A box contains **10 cards**, consisting of **6 red cards and 4 blue cards**. One of the following two procedures is selected at random:

- **Procedure A:** 3 cards are selected randomly **without replacement**.
- **Procedure B:** 3 cards are selected randomly **with replacement**.

After the procedure is completed, it is observed that **exactly 2 of the 3 selected cards are red.**

What is the probability that **Procedure A** was used? [10 points]

Q2. A person tells the truth **2 out of every 3 times**. Another person, who tells the truth **3 out of every 4 times**, is randomly selected to verify the statement. A particular statement is asked to the first person **5 times independently**. It is observed that he gives the **correct answer exactly 3 times**. What is the probability that the person was **telling the truth on the first occasion**, given that he gave exactly 3 correct answers in the 5 occasions? [10 points]

Q3. Demonstrate the training of the naïve baye's classifier for the given dataset.

Student	Study Time	Attendance	Assignment	Class
S1	High	High	Complete	Excellent
S2	High	High	Complete	Excellent
S3	High	Low	Complete	Excellent
S4	Low	High	Complete	Average
S5	High	Low	Incomplete	Average
S6	Low	High	Incomplete	Average
S7	Low	Low	Complete	Poor
S8	Low	Low	Incomplete	Poor
S9	Low	Low	Incomplete	Poor

Now given a test sample as Study Time = High, Attendance = Low, Assignment = Complete. Find the category of the test sample using naïve bayes classifier. [10 points]

CSE 32

CT:03 Class Roll#_____ Name _____ Full Marks: 25 Time: 40 minutes

Q1. Given the loan approval dataset as follows:

Age Group	Income Group	Credit History	Loan Approved
Young	Low	Good	Yes
Young	High	Poor	No
Middle-Aged	Low	Poor	No
Middle-Aged	High	Good	Yes
Senior	Low	Good	Yes
Senior	High	Good	Yes
Young	Low	Poor	Yes
Middle-Aged	Low	Good	Yes
Senior	High	Poor	No

Now answer the following questions:

i. Compute the prior probabilities of ***Loan Approved*** feature. [2 points]

ii. Given the following feature vector of a user as:
X = [Young, High, good]
Predict if the user's loan is approved? [5 points]

iii. Let X be the random variable to represent the number of times a loan is approved.
Compute E[X]. [2 points]

Q2. Mention the difference between covariance and correlation. Derive the formula for the Pearson correlation coefficient. [5 points]

Q3. Give the concept of stochastic gradient descent. For the following 2-input AND gate dataset, show the computation for SGD for one epoch [taking]. Mention advantage and disadvantage of SGD over batch gradient descent. [5 points]

X_1	X_2	Y
0	0	0
0	1	0
1	0	0
1	1	1

Q4. In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals:

	Moderate Smokers	Heavy Smokers
Nonsmokers	21 (H), 48 (NH)	36 (H), 30 (NH)
Moderate Smokers	26 (H), 26 (NH)	19 (H), 19 (NH)

where **H** and **NH** in the table stand for **Hypertension** and **Nonhypertension**, respectively. If one of these individuals is selected at random, find the probability that the person is:

(a) experiencing hypertension, given that the person is a heavy smoker. [3 points]

(b) a nonsmoker, given that the person is experiencing no hypertension. [3 points]

Another Question

Course Title: Introduction to probability and statistics Course Code: CSE 251 Full Marks: 30

Name: _____ Roll: _____ Year _____ Semester _____

(দেখি সব পারি কি না)

Q1. A data scientist compares the **box plots of execution times** (in milliseconds) of four search algorithms: **Linear Search (LS)**, **Binary Search (BS)**, **Jump Search (JS)**, and **Exponential Search (ES)**.

Metric	LS	BS	JS	ES
Q1	40	10	18	8
Median	55	14	24	10
Q3	70	18	30	12
Max	200	22	36	14
Min	30	8	15	7

a) Which algorithm has the most consistent (stable) performance? Explain. [3 points]

b) Which algorithm is likely best suited for real-time applications requiring low-latency (time required to produce the output) and stable performance? Explain. [3 points]

c) Find if there is any outlier in LS? If yes, what could be the possible interpretation of the performance of LS algorithm? [4 points]

Q2. Suppose the **maximum heart rate (thalach)** percentiles of patients from a heart disease dataset are calculated as:

- ✓ P25: 120 bpm
- ✓ P50: 150 bpm
- ✓ P75: 170 bpm

A cardiologist wants to flag extreme heart responses.

Suggest a data-driven rule to define what might be considered **abnormally high** or **abnormally low** heart rates. [3 points]

Q3. The mean cholesterol level in the dataset is **250 mg/dL**, and the median cholesterol level is **220 mg/dL**. What does this difference suggest about the cholesterol data distribution? Which measure (mean or median) is more appropriate to report to a physician for typical cholesterol levels? Explain.

[3 points]

Q4. Given the following cholesterol values for 15 patients:

[155, 160, 165, 170, 185, 190, 195, 200, 210, 220, 230, 250, 260, 280, 295]

Draw the histogram. [6 points]

Q5. Given two datasets as follows:

Dataset-1	*Dataset-2*
A cybersecurity team wants to predict how long an employee account will remain uncompromised, given columns: ✓ password_length (integer) ✓ num_prev_phishing_clicks (integer) ✓ department (let's say A, B, C etc.) ✓ 2FA_enabled (0/1) ✓ last_password_change_days (days) ✓ days_to_breach (days)	An email spam dataset columns: ✓ Email_Text (raw text) ✓ Sender_Domain (textual data) ✓ Number_of_Links (integer) ✓ Contains_Images (boolean) ✓ Spam (0 = not spam, 1 = spam)

Now answer the following for both datasets: [6 points]

a) Is this a classification or regression problem?
b) Identify the target variable and feature variables.
c) Which features are numeric and which are categorical?
d) What type of variable is the target?

Q6. Define outlier, population, and sample. [2 points]

Final Question

Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 2nd Semester B.Sc. (Hons.) Final Examination -2024
Course Title: Introduction to probability and statistics

Time: 3:00 hours | Course Code: STAT 251 | Full Marks: 60

[Answer all the sections. Figures in the right margin indicate marks]

Q1. ***Section-I (CO1): Answer all the questions.***

2x6 =12

a) State and prove Bayes' theorem.
b) Outline the concept of data, dataset, and data distribution.
c) Give the notion of sample, population, and confidence interval.
d) State the central limit theorem.
e) Define z-score.
f) List out the differences between the Bernoulli distribution and the binomial distribution.

Q2. ***Section-II (CO2): Answer any three out of the following questions.***

a) A data scientist compares the **box plots of execution times** (in milliseconds) of four search algorithms: **Linear Search (LS)**, **Binary Search (BS)**, **Jump Search (JS)**, and **Exponential Search (ES)**. [4]

Metric	LS	BS	JS	ES
Q1	40	10	18	8
Median	55	14	24	10
Q3	70	18	30	12
Max	200	22	36	14
Min	30	8	15	7

i) Explain which searching algorithm has the most consistent performance.
ii) Identify the algorithm that is likely to be best suited for real-time applications requiring low-latency (time needed to produce the output) and stable performance.
iii) Find if there is any outlier in LS? If yes, what could be the possible interpretation of the performance of the LS algorithm?

b) The mean cholesterol level in the dataset is 250 mg/dL, and the median cholesterol level is 220 mg/dL. Interpret what this difference suggests about the cholesterol data distribution. Choose a more appropriate measure between the mean and the median to report to a physician for typical cholesterol levels. [4]

c) A random sample of 100 recorded deaths in Bangladesh during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, discuss if this seems to indicate that the mean life span today is greater than 70 years. Use a 0.05 level of significance. [4]

d) A nutrition researcher wants to predict the calorie burn per workout session (y, in hundreds) from the exercise duration (x, in tens of minutes). He collects data from 5 training sessions:

X	1	2	3	4	5
y	1.8	2.4	3.2	4.1	5.0

[4]

From this dataset, he designed the best-fit regression model as $\hat{y} = 0.9 + 0.8X$. Now determine the coefficient of determination R^2 and interpret the result.

Q3. ***Section-III (CO3): Answer any three out of the following questions.***

a) A company tracks weekly advertising spend and weekly sales:

Week	Ad Spend (X)	Sales (Y)
1	100	500
2	120	550
3	140	600
4	160	650

i) Compute the correlation between Ad Spend and Sales.
ii) Create a covariance matrix, a heat map, and interpret them.

b) A certain brand of upright freezer is available in three different rated capacities: 16 ft^3, 18 ft^3, and 20 ft^3. Let X=the rated capacity of a freezer of this brand sold at a certain store. Suppose that X has the following pmf: [4]

x	16	18	20
$p(x)$	0.2	0.5	0.3

i) If the price of a freezer having capacity X is $70X-650$, determine the expected price paid by the next customer to buy a freezer.
ii) Suppose that although the rated capacity of a freezer is X, the actual capacity is

$$h(X) = X^2 - 0.008X^2$$

Compute the expected actual capacity of the freezer purchased by the next customer.

c) Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as: [4]

$f(x,y)$		x		
		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

i) Evaluate the marginal distribution of X and Y.
ii) Determine $P(Y = 3 \mid X = 2)$.

d) A large industrial firm purchases several new word processors at the end of each year. The number of units purchased depends on how often repairs were needed during the year. Let the random variable X denote the number of word processors purchased at the year's end. The probability distribution of X is given below: [4]

x	0	1	2	3
$f(x)$	1/10	3/10	2/5	1/5

If the cost of the desired model is BDT 1200 per unit and at the end of the year a refund of $50X^2$ (BDT) will be issued, calculate how much money the firm expect to spend on new word processors during this year.

Q4. ***Section-IV (CO4): Answer any three out of the following questions.***

a) A survey conducted in Bangladesh by public health researchers found that 60% of Diazepam users first took the drug for psychological reasons. [4]
If 8 users from different districts are selected and interviewed at random, analyze the distribution of survey records and answer the following:
i) Estimate the probability that exactly 3 began using Diazepam for psychological reasons.
ii) Calculate the probability that at least 5 began using it for non-psychological reasons.

b) The IQs of 600 applicants to a certain college are approximately normally distributed with a mean of 115 and a standard deviation of 12. If the college requires an IQ of at least 95, investigate how [4]

many students will be rejected on the basis of IQ, regardless of their qualifications. Note that IQs are recorded to the nearest integers.

c) The error involved in making a certain measurement is a continuous random variable X with pdf [4]

$$f(x) = k(4 - x^2) \quad if \; -2 < x < 2$$

i) Find the value of k for which ***f(x)*** is a probability density function.
ii) Compute $P(X < -0.5 \; or \; X > 0.5)$.

d) A manufacturing firm produces steel pipes in three plants with daily production volumes of 500, 1000 and 2000 units respectively. According to the past experience, it is known that the fraction of the defective outputs produced by the three plants are respectively 0.005, 0.008 and 0.10. If a pipe is selected from a day's total production and found to be defective, find out [4]
i) From which plant the pipe comes?
ii) Assess the probability that it came from the first plant.

Q5. ***Section-V (CO5): Answer any two out of the following questions.***

a) A chain of video stores sells three different brands of DVD players. Of its DVD player sales, 50% are brand 1 (the least expensive), 30% are brand 2, and 20% are brand 3. Each manufacturer offers a 1-year warranty on parts and labor. It is known that 25% of brand 1's DVD players require warranty repair work, whereas the corresponding percentages for brands 2 and 3 are 20% and 10%, respectively. [6]

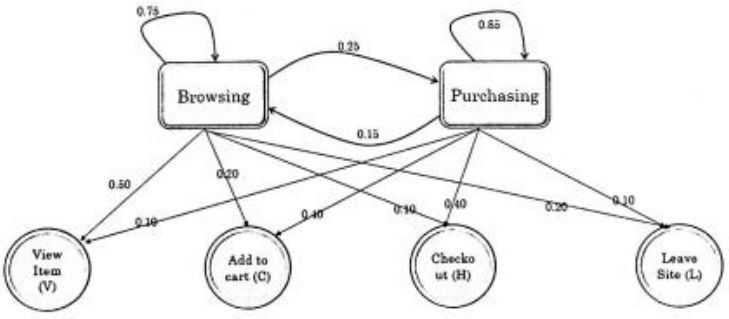
iii) What is the probability that a randomly selected purchaser has bought a brand 1 DVD player that will need repair while under warranty?
iv) What is the probability that a randomly selected purchaser has a DVD player that will need repair while under warranty?
v) If a customer returns to the store with a DVD player that needs warranty repair work, what is the probability that it is a brand 1 DVD player? A brand 2 DVD player? A brand 3 DVD player?

b) The grades of a class of 6 students on a midterm report (x) and on the final examination (y) are as follows: [6]

x	77	50	71	72	81	94
y	82	66	78	34	47	85

Simulate the computation of Stochastic Gradient Descent (SGD) for one epoch. Choose $\eta = 0.1$.

c) A retail analytics team models customer shopping behavior using a Hidden Markov Model (HMM) with two hidden states and four observed states as follows: [6]



Given the following observed action sequence: X=[V,C,H]

Use maximum likelihood reasoning to predict whether the sequence is more likely generated from a customer who is mostly Browsing or mostly Purchasing.

Annex-I **(Z-score Table)**

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

CSE 31

Department of Computer Science and Engineering

STAT 251: Introduction to Probability and Statistics

Class Test 2

Duration: 30 mins Full Marks: 25

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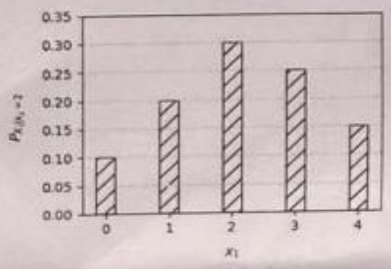
1. Answer any ***three*** of the following ***five*** questions.

(a) Define: (i) Joint Probability Distribution (ii) Covariance. (5)

(b) Given a probability distribution $P(x_1, x_2)$. Find $P_1(x_1)$, $P_2(x_2)$ and $P_{X_2|x_1<2}(x_2)$. (5)

$x_1 \backslash x_2$	0	1	2
0	0.1	0.2	0.1
1	0.05	0.15	0.1
2	0.1	0.05	0.15

(c) Consider the conditional probability distribution graph. Find $E(X_1|x_2 = 2)$. (5)



(d) Six independent space mission to the moon are planned. The estimated probability of success on each mission if 0.95. Find the probability that at least six of the planned mission will be successful. (5)

(e) A potential customer enters an automobile dealership every hour. The probability of a salesperson concluding a transaction is 0.10. She is determined to keep working until she has sold three cars. What is the probability that she will have to work exactly 8.5 hours or more? (5)

2. Answer any ***one*** of the following ***two*** questions.

(a) A refrigerator manufacturer subjects his finished probducts to a final inspection. Of interest are two categories of defects: scratches and flaws in the porcelain finish, and mechanical defects. The number of each type of defect is a random variable. The results of inspecting 50 refrigerators are show in the following table, where X represents the occurence of finish defects and Y represents the occurence of mechanical defects. (10)

$Y \backslash X$	0	1	2	3	4	5
0	11/50	3/50	2/50	1/50	1/50	1/50
1	8/50	3/50	3/50	1/50	1/50	
2	4/50	3/50	2/50	1/50		
3	3/50	1/50				
4	1/50					

(i) Find the marginal distributions of X and Y.

(ii) Find the probability distribution of mechanical defects given that there are no finish defects.

(iii) Find the probability distribution of finish defects given that there are no mechanical defects.

(b) The XYZ Company plans to visit potential customers until a substantial sale is made. Each sales presentation costs 1000 USD. It costs 4000 USD to travel to the next customer and set up a new presentation. (10)

(i) What is the expected cost of making a sale if the probability of making a sale after any presentation is 0.10?

(ii) If the expected profit from each sale is 15,000 USD, should the trips be undertaken?

(iii) If the budget for advertising is only 100,000 USD, what is the probability that this sum will be spent without getting an order.

GOOD LUCK!

JAHANGIRNAGAR UNIVERSITY

Department of Computer Science and Engineering

STAT 251: Introduction to Probability and Statistics

Class Test 3

Duration: 30 mins Full Marks: 25

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1. Answer any ***three*** of the following ***five*** questions.

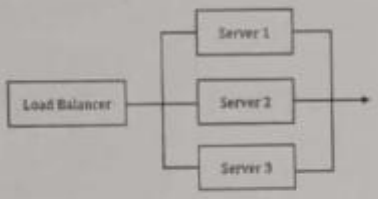
   (a) Define with density plots: (i) Uniform Distribution (ii) Weibull Distribution. (5)

   (b) State the memory less property of the exponential distribution. (5)

   (c) Is there an exponentially density that satisfies the following condition? If so, find the value of λ. (5)

   $$P\{X \le 2\} = \frac{2}{3} P\{X \le 3\}$$

   (d) A box of candy contains 24 bars. The time between demands for these candy bars is exponentially distributed with a mean of 10 minutes. Find the probability that a box of candy bars opened at 8:00AM will be empty by 12:00PM. (5)



   (e) Consider a social media system with a load balancer and three independent servers given above. Initially Server 1 is online, and Server 2 and 3 are on standby. If Server 1 fails, the load balancer switches to Server 2 on until it fails and then Server 3 is switched on. If the system lifespan in hours follows a gamma distribution with $r = 3$, $\lambda = 0.01$, find the reliability of the system ***R(x = 500 hours)*** i.e. no failures happen within 500 hours. (5)

2. Answer any ***one*** of the following ***two*** questions.

   (a) The percentage of cotton in a material used to manufacture men's shirts is give below. Construct a histogram for the data. Comment on the properties of the data. (10)

34.2	33.6	33.8	34.7	37.8	32.6
33.1	34.7	34.2	33.6	36.6	35.1
34.5	35.0	33.4	32.5	35.4	34.6
35.6	35.4	34.7	34.1	34.6	35.9
34.3	33.2	34.6	35.1	33.8	34.7
35.1	36.8	35.2	36.8	37.1	33.6

   (b) Compute the approximate sample mean, sample variance, median, and mode from the data in the following frequency distribution: (10)

Class Interval	Frequency
$-10 \le x < 0$	3
$0 \le x < 10$	8
$10 \le x < 20$	12
$20 \le x < 30$	16
$30 \le x < 40$	9
$40 \le x < 50$	4
$50 \le x < 60$	2

Final Question

Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 2nd Semester B.Sc. (Hons.) Examination 2023

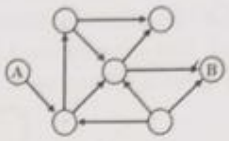
Course Title: **Introduction to Probability and Statistics** Course No: **STAT 251**
Time: **3 hours** Full Marks: **60**

[Answer ***any six*** out of ***eight*** of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

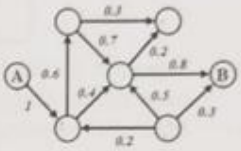
1. a. Differentiate between discrete and continuous random variables with examples. *2*

   b. A player starts a maze from point ***A***. Determine the probability that the player arrives at point ***B*** for the following two cases. *3*

   Case 1. At any point the player selects the next point randomly.

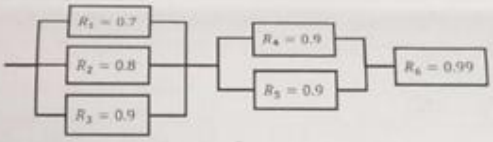


   Case 2. At any point the probability to select the next point is given.

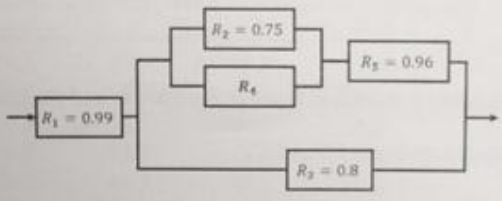


   c. Consider a software company has two data centers. Each data center consists of several servers but may have different network configurations. If each server works independently. *5*

   (i) Find the reliability of the Data Center 1.



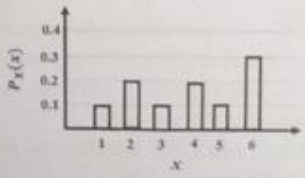
   (ii) Determine the required reliability R_4 in order to maintain the reliability of the Data Center 2 at least ***0.9***.



2. a. State Chebyshev's Inequality. *2*

   b. Consider the probability distribution $P_X(x)$ of a discrete random variable X. *3*

   (i) Find $E(X)$.
   (ii) Draw the CDF $F_X(x)$ for $P_X(x)$.
   (iii) Find $P_X(2 < x \leq 4)$ *using the CDF.*



   c. A continuous random variable X has a density function *5*

$$f_X(x) = \frac{2x}{9}, \quad 0 < x < 3$$
$$= 0 \quad \textit{otherwise.}$$

   (i) Develop the $F_X(x)$.

(ii) Find the mean of X and the variance of X.
(iii) Find the value m such that $P_X(X \geq m) = P_X(X \leq m)$.

3. a. If $B_1, B_2, B_3, \ldots, B_k$ represents a partition of sample space S and A is an arbitrary event on S, then state the total probability of A with a Venn diagram. 2

b. Consider the probability distribution $P_X(x)$ of a random variable X as tabulated and a function $H(X = x) = x^2 + 1$. 3

x	-3	-2	-1	0	1	2	3
$P_X(x)$	0.05	0.1	0.2	0.3	0.2	0.1	0.05

(i) Find the probability distribution of $P_Y(y)$ where $Y = H(X)$.
(ii) Determine $E[H(x = 0)]$.

c. A software company has developed a new password authentication system. To set a password you need to follow the rules below. 5

Rule 1. The password consists of 4 digits.
Rule 2. You can only use *, #, 1, 2, A, B as characters.
Rule 3. No character can be used more than once.



(i) Determine the number of unique passwords using the multiplication principle.
(ii) Draw the tree diagram showing the passwords that start with '*'.
(iii) Determine the probability that the password ends with '#' if Rule 3 is not followed.

4. A refrigerator manufacturer subjects his finished products to a final inspection. Of interest are two categories of defects: scratches or flaws in the porcelain finish and mechanical defects. The number of each type of defect is a random variable. The results of inspecting 50 refrigerators are shown in the following table where X represents the occurrence of finish defects and Y represents the occurrence of mechanical defects.

Y \ X	0	1	2	3	4	5
0	11/50	4/50	2/50	1/50	1/50	1/50
1	8/50	3/50	2/50	1/50	1/50	
2	4/50	3/50	2/50	1/50		
3	3/50	1/50				
4	1/50					

a. Define joint probability distribution function for discrete and continuous random variables. 3
b. Find the marginal distribution of X and Y. 2
c. Find the probability distribution of mechanical defects given that there are no finish defects. 2.5
d. Find the probability distribution of finish defects given that there are no mechanical defects. 2.5

5. A lot of 25 color television tubes are subjected to an acceptance testing procedure. The procedure consists of drawing five tubes at random, without replacement, and testing them. If two or fewer tubes fail, the remaining ones are accepted. Otherwise, the lot is rejected. Assume the lot contains four defective tubes.

a. Define Binomial distribution with mean and variance. State an application of the distribution. 3
b. Calculate the exact probability of lot acceptance? 3
c. Calculate the probability of lot acceptance computed from the binomial distribution with $p = \frac{4}{25}$. 4

6. a. Derive the equation for the mean and variance of the Poisson distribution and state an example where it is applicable in engineering. *4*

b. The life of an electronic system is $Y = X_1 + X_2 + X_3 + X_4$, the sum of the subsystem lives. The subsystems are independent, each having exponential failure densities with a mean time between failures of 4 hours. Find the probability that the system will operate for at least 24 hours. *3*

c. Consider the following condition $P\{X \leq 2\} = \frac{2}{3}P\{X \leq 3\}$. Determine if there any exponential density that satisfies the condition. *3*

7. a. Mention real-world applications of statistical regression models. *2*

b. An article in Computers and Industrial Engineering describes the time-to-failure data (in hours) for jet engines. Some of the data are reproduced below. *8*

Engine No	Failure Time	Engine No	Failure Time
1	150	14	171
2	291	15	197
3	93	16	200
4	53	17	262
5	2	18	255
6	65	19	286
7	183	20	206
8	144	21	179
9	223	22	232
10	197	23	165
11	187	24	155
12	197	25	203
13	213		

(i) Construct a frequency distribution and histogram for these data.

(ii) Calculate the sample mean, sample median, sample variance, and sample standard deviation.

8. a. Briefly describe the measure of central tendency. *2*

b An article in *Technometrics* presents the following data on motor fuel octane ratings of several blends of gasoline. *8*

88.5	87.7	83.4	86.7	87.5
95.6	93.3	94.7	91.1	91.0
88.3	87.6	84.3	86.7	88.2
94.2	92.7	93.2	91.0	90.3
89.2	88.3	85.3	87.9	88.6

(i) Construct a stem-and-leaf plot.

(ii) Find the sample median and sample mode.

(iii) Determine the skewness and kurtosis measures.

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GOOD LUCK!

CSE 30

Jahangirnagar University
Department of Computer Science and Engineering
2nd Tutorial, Course Title: Introduction to Probability and Statistics, Course Code: STAT-251

Marks: 15 **Time: 30 Minutes**

1. (a) Define Conditional Probability and Bayes Theorem.

(b) Three printers do work for the publications office of Georgia Tech. The publications office does not negotiate a contract penalty for late work, and the data below reflects a large amount of experience with these printers.

Printer i	Fraction of Contracts Held by Printer i	Fraction of Deliveries More than One Month Late
1	0.2	0.2
2	0.3	0.5
3	0.5	0.3

A department observes that its recruiting booklet is more than a month late. What is the probability that the contract is held by printer 3?

(c) Suppose that the joint density density of $[X_1, X_2]$ is given by

$$f(x_1, x_2) = 6x_1, \quad 0 < x_1 < x_2 < 1$$
$$= 0, \quad \text{otherwise.}$$

i) Find the marginal distributions of X_1, and X_2.
ii) Calculate the conditional distribution of X_1 on X_2.
iii) Calculate the correlation coefficient.

Jahangirnagar University
Department of Computer Science and Engineering
1st Tutorial, Course Title: Introduction to Probability and Statistics, Course Code: STAT-251

Marks: 25 **Time: 90 Minutes**

1. Ultrasound was used to gather the accompanying corrosion data on the thickness of the floor plate of an aboveground tank used to store crude oil ("Statistical Analysis of UT Corrosion Data from Floor Plates of a Crude Oil Aboveground Storage Tank," *Materials Eval.*, 1994: 846-849; each observation is the largest pit depth in the plate, expressed in milli-in.

90 125 100 115 90 95 92 125 98 85 60 94 55 52 94 85 40 75 70

i) Construct a histogram and comment on the properties of the data.
ii) What proportion of the observations are less than 100?
iii) Construct a box plot for the data. Interpret data using this plot.
iv) Compute the sample mean and sample standard deviation from the frequency distribution, hence comment on the findings.

2. The article "Photocharge Effects in Dye Sensitized Ag[Br,I] Emulsions at Millisecond Range Exposures" (*Photographic Sci. and Engr.*, 1981: 138-144) gives the accompanying data on x = % light absorption at 5800 A and y = peak photovoltage.

x	4.0	8.7	12.7	19.1	21.4	24.6	28.9	29.8	30.5
y	0.12	0.28	0.55	0.68	0.85	1.02	1.15	1.34	1.29

i) Construct a scatter plot of this data. What does it suggest?
ii) Fit a simple linear regression model to the data.
iii) Predict peak photovoltage for 2% light absorption.
iv) Test for significance of regression at 5% level of significance.
v) Determine coefficient of variation and hence comment on it.

3. Multiple Regression Analysis: Pull Strength (y) versus Wire Length (x_1), Die Height (x_2)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	2	5990.8	2995.39	572.17	0.000
Wire Length (x1)	1	4507.5	4507.53	861.01	0.000
Die Height (x2)	1	104.9	104.92	20.04	0.000
Error	22	115.2	5.24		
Total	24	6105.9			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2.28805	98.11%	97.94%	97.44%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	2.26	1.06	2.14	0.044	
Wire Length (x1)	2.7443	0.0935	29.34	0.000	1.17
Die Height (x2)	0.01253	0.00280	4.48	0.000	1.17

Fits and Diagnostics for Unusual Observations

Obs	Pull Strength (y)	Fit	Resid	Std Resid	
15	21.650	15.809	5.841	2.65	R
17	69.000	64.666	4.334	2.20	R

Figure: Minitab Output

i) Write down the fitted regression equation.
ii) Does the chosen model appear to specify a useful relationship between the dependent variable and the predictors?
iii) Interpret the estimated slope and the coefficient of determination.

Final Question

Department of Computer Science and Engineering
Jahangirnagar University
2nd Year 2nd Semester B.Sc. (Hons.) Final Examination 2022
Course Title: Introduction to Probability and Statistics
Course No. STAT 251

Full Marks: 60 **Time: 03 Hours**

Answer any **Six (06)** of the following questions. Each question carries equal marks.

1. (a) Define the term Statistics, Random Variable, Qualitative data, Quantitative data, and Five summary measures with suitable examples. 5
   (b) Explain the terms Correlation, point Biserial correlation, and Rank correlation. Briefly describe in what situation these correlations are used. Are these types of correlations important in your field of study? Briefly discuss. 5

2. Ultrasound was used to gather the accompanying corrosion data on the thickness of the floor plate of an aboveground tank used to store crude oil ("Statistical Analysis of UT Corrosion Data from Floor Plates of a Crude Oil Aboveground Storage Tank," Materials Eval., 1994: 846-849; each observation is the largest pit depth in the plate, expressed in milli-in. 10

90 125 100 115 90 95 92 125 98 85 60 94 55 52 94 85 40 75 70

   i) Construct a frequency distribution for the given data.
   ii) What proportion of the observations are less than 100?
   iii) Construct a histogram and comment on the properties of the data.
   iv) Compute the sample mean and sample standard deviation from the frequency distribution, hence comment on your findings.

3. The article "Photocharge Effects in Dye Sensitized Ag[Br,I] Emulsions at Millisecond Range Exposures" (*Photographic Sci. and Engr.*, 1981: 138-144) gives the accompanying data on x = % light absorption at 5800 A and y = peak photovoltage. 10

% light absorption	4.0	8.7	12.7	19.1	21.4	24.6	28.9	29.8	30.5
Peak voltage	0.12	0.28	0.55	0.68	0.85	1.02	1.15	1.34	1.29

   i) Construct a scatter plot of this data. What does it suggest?
   ii) Fit a simple linear regression model to the data.
   iii) Predict peak photovoltage for 2% light absorption.
   iv) Test for significance of regression at 5% level of significance.
   v) Determine the coefficient of determination and hence comment on it.

4. (a) Define Conditional Probability and Bayes Theorem. State and prove the Chebyshev's Inequality. 5

   (b) Three printers do work for the publications office of Georgia Tech. The publications office does not negotiate a contract penalty for late work, and the data below reflect a large amount of experience with these printers. 5

Printer i	Fraction of Contracts Held by Printer i	Fraction of Deliveries More than One Month Late
1	0.2	0.2
2	0.3	0.5
3	0.5	0.3

A department observes that its recruiting booklet is more than a month late. What is the probability that the contract is held by printer 3?

5. (a) Define Binomial distribution. State some of its important properties. Show that Poisson distribution is a limiting form of binomial distribution. 5

(b) Find the mean and variance of Poisson distribution. 5

6. (a) Briefly discuss the Exponential distribution, Gamma distribution, Normal distribution, and Weibull distribution. Write down some applications of Normal distribution. 5

(b) A production process produces thousands of parts per day. On average, 1% of the parts are defective and this average does not vary with time. Every hour, a random sample of 100 parts is selected from a conveyor, and several characteristics are observed and measured on each part; however, the inspector classifies the part as either good or defective. 5

i) Find the mean number of defective parts.
ii) What is the probability that the sample has more than two defectives?

7. (a) Define the Central Limit Theorem. How large must n be to get reasonable results using the normal distribution to approximate the distribution of Y? 5

(b) The breaking strength (in newtons) of a synthetic fabric is denoted X, and it is distributed as $N(800, 144)$. The purchaser of the fabric requires the fabric to have a strength of at least 772 nt. A fabric sample is randomly selected and tested. What is the probability that it will fulfill the purchaser's requirement? 5

8. (a) Define the Stochastic Process with a suitable example. What are the different types of Stochastic Processes? 5

(b) Suppose a machine can be labeled as state 0, 1, and 2 representing that the machine is idle, malfunctioning, and running respectively. The state of the system for different days are: 1-0-1-1-0-2-0-1-2-2-1-0-0-1-1-1-1-1-2-1-2-0-1-1-2-1-2-0-2-0-2-1-1-2-0-1-2-1-1-0-2-0-2-1-1-1-2-2-0. 5

i) Find the transition matrix and transition probability matrix.
ii) Draw the transition probability graph.
iii) Under usual notations find the value of $p_{01}^{(1)}$, $p_{22}^{(1)}$, $p_{21}^{(2)}$, $p_{21}^{(3)}$.